

No. of Pages	2
No. of Questions	6

Department of CSE, BRAC University

CSE321:Operating Systems

Make-up Final Exam, Spring 2019

Time: 2hours

Full Marks: 60

Answer any Two from Section - A (2 X 10 = 20)

Section - A

1. a. What OS's Do? A modern computer system has a number of device controllers to do specific task; what is that task? [4]
b. In multithreaded programs how can you cancel a thread and maintain a thread Pool. [2]
c. What are differences between:
i. Process and Thread (with figure). [2]
ii. Asymmetric and symmetric multiprocessing system. [2]
2. Suppose, we have the following scenario in an OS. There are four processes and four resource types. Answer the following questions using Banker's Algorithm. [10]

Processes	Max				Allocation				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
Pa	7	0	1	3	7	0	0	2	1	4	4	4
Pb	2	7	5	0	2	1	0	0				
Pc	2	12	5	6	0	6	3	3				
Pd	1	6	5	6	0	2	1	2				

- a. Calculate Need matrix.
- b. Is this system in safe state? If yes, then find the safe sequence or if not, then provide necessary explanation?
- c. What happens if process Pb request at this moment for (0, 4, 2, 0)? Whether Banker's algorithm grants the request or not? If grants the request find the safe sequence.
3. a. What aging is? How multilevel feedback queue work? What are the optimization criteria of process scheduling? [5]
b. What are the difference between Internal and External Fragmentation? Name three requirements for solving critical section problem. [5]
4. a. How to predict the next CPU burst time of the process for SJF scheduling algorithm? [2]
b. Considering following set of processes, construct Gantt Chart for SJF (preemptive) and RR (time quantum = 3 milliseconds) and Priority scheduling (a smaller priority number implies a higher priority) CPU scheduling algorithm. [8]

*** Note: Need to consider Arrival time.

Process	Burst Time	Arrival Time	Priority
P1	10	0	1
P2	13	13	2
P3	9	15	3
P4	8	20	5
P5	5	50	7

P6	12	90	9
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SECTION – B(20 marks)

6. a. Assume that page size = 4 bytes and Physical Memory = 32 bytes. If CPU generates logical addresses 8, 3, 13, 0, 17 and 7 respectively then how the users' view of memory can be mapped into physical memory? [6]

P0	I
P1	Love
P2	CSE
P3	And
P4	BRAC
P5	University

Logical Memory

P0	7
P1	9
P2	4
P3	8
P4	5
P5	6

PMT

Main Memory

- b. a. A very simple computer has 4 page frames. And a process makes the following list of page references: 1, 2, 3, 4, 1, 5, 2, 3, 1, 2, 3, 4, 5, 1, 2, 5, 4, 3, 3, 1, 0, 4, 3, 1, 2. [6]
- c. How many page fault occur using LRU and Optimal page replacement algorithms? [3]
- d. When does page fault occur? Draw a figure to illustrate how Shared Pages (sharing of common code) work in paged memory management system. [5]
- e. a. Given memory partitions of 210k, 350k, 250k, 190k, 250k and 452k (in order, top to bottom), apply first fit and best fit algorithms to place processes with the space requirement of 250k, 425k, 212k, 160, 210, and 440k (in order). Which algorithm makes the most effective use of memory?

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